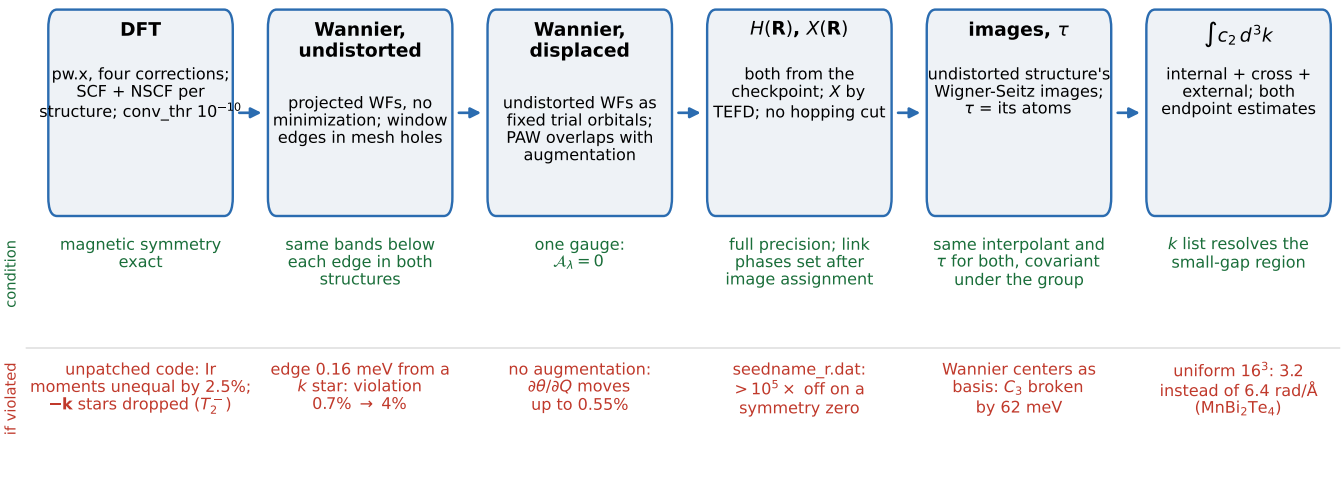
DFT
pw.x, four corrections; SCF + NSCF per structure; conv_thr 10^{-10}
Wannier, undistorted
projected WFs, no minimization; window edges in mesh holes
Wannier, displaced
undistorted WFs as fixed trial orbitals; PAW overlaps with augmentation
$H(\mathbf{R})$, $X(\mathbf{R})$
both from the checkpoint; X by TEFD; no hopping cut
images, τ
undistorted structure's Wigner-Seitz images; τ = its atoms
$\int c_2\, d^3k$
internal + cross + external; both endpoint estimates
condition
magnetic symmetry exact
same bands below each edge in both structures
one gauge: $\mathcal{A}_\lambda = 0$
full precision; link phases set after image assignment
same interpolant and τ for both, covariant under the group
k list resolves the small-gap region
if violated
unpatched code: Ir moments unequal by 2.5%; $-\mathbf{k}$ stars dropped (T_2^-)
edge 0.16 meV from a k star: violation 0.7% → 4%
no augmentation: $\partial\theta/\partial Q$ moves up to 0.55%
seedname_r.dat: $> 10^5\times$ off on a symmetry zero
Wannier centers as basis: C_3 broken by 62 meV
uniform 16^3: 3.2 instead of 6.4 rad/Å ($MnBi_2Te_4$)